

Group Name: Waystar



Experiment 1

1. Resistor



Resistance Value

1st Digit = Orange = 3

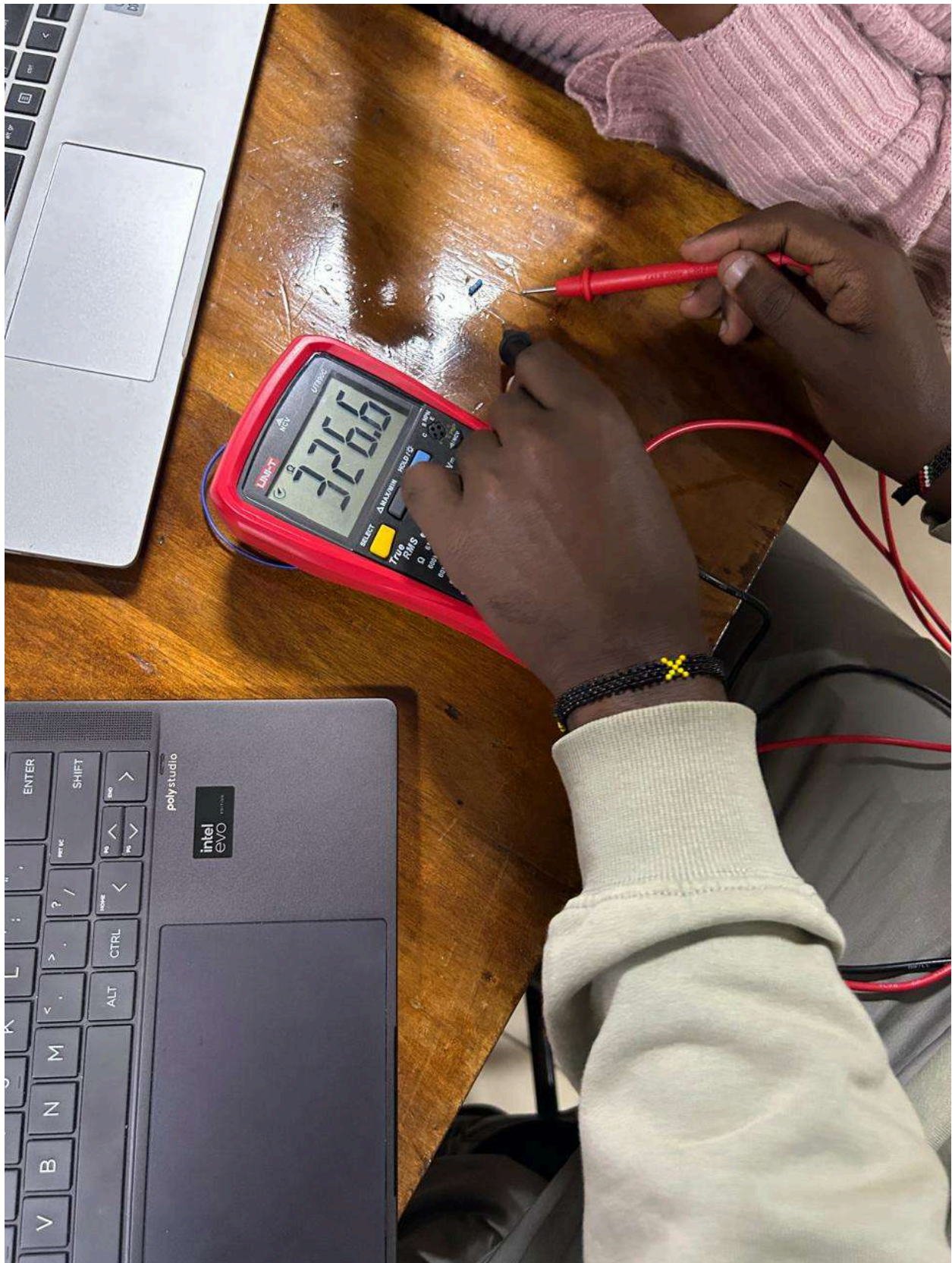
2nd Digit = Orange = 3

X = Brown = 10^1

Tolerance = Gold = 5%

Therefore, resistance is $330\Omega \pm 5\%$

2. Multimeter Resistance Value = $326.6\ \Omega$

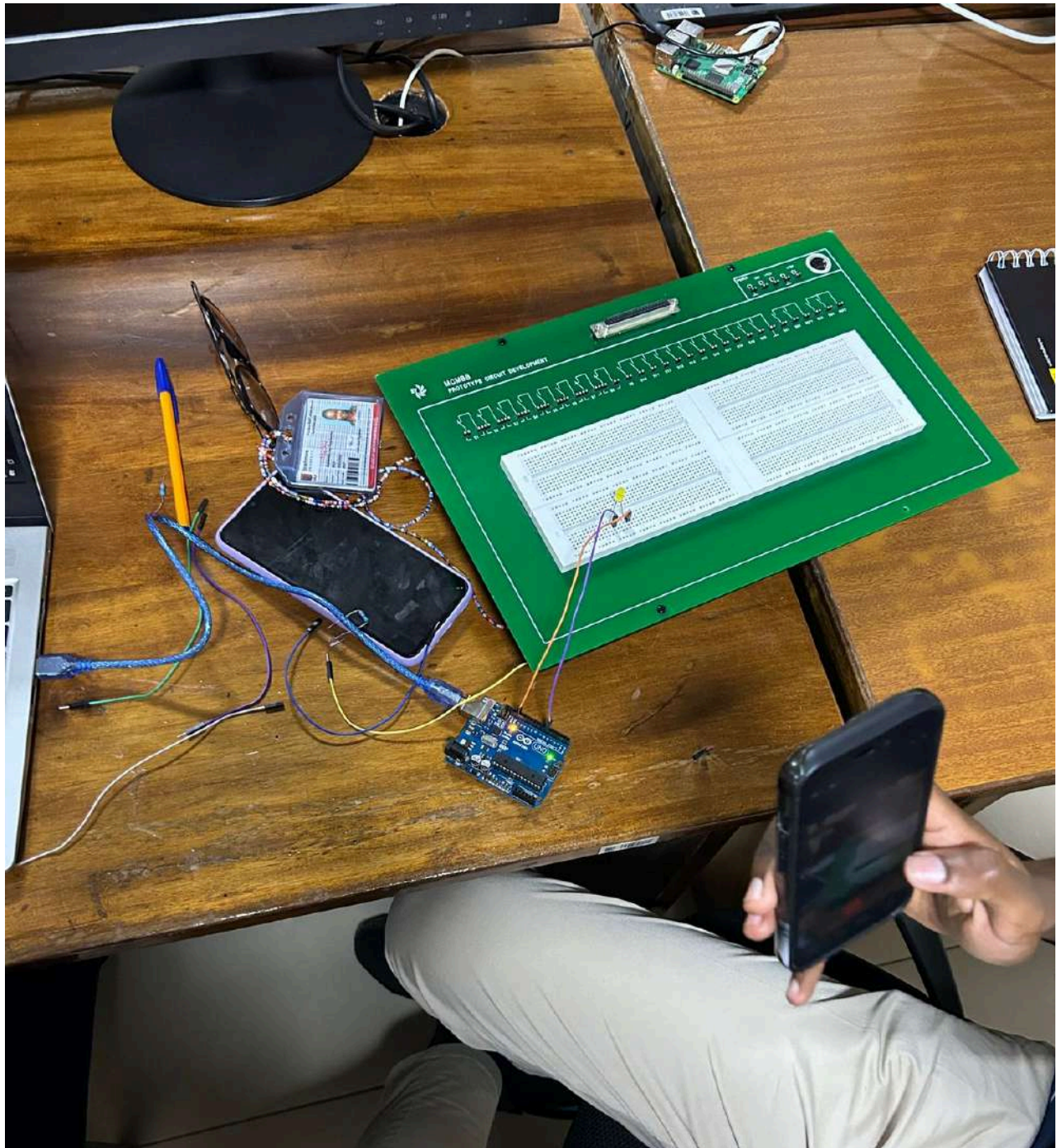


Where does the difference in resistance values arise from? Differences arise from manufacturing tolerances ($\pm 5\%$ in our case), operating temperature changes, and measurement errors.

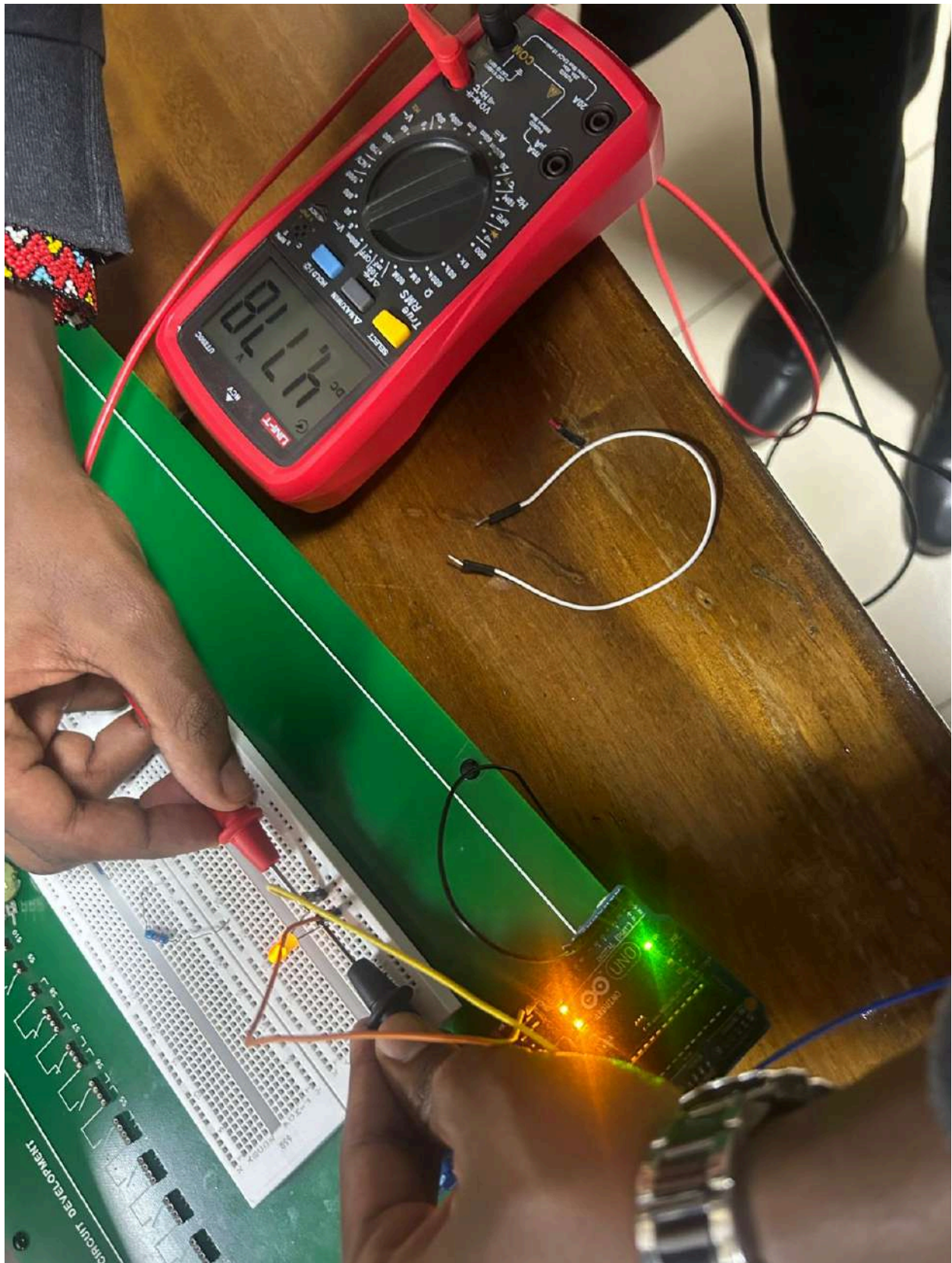
Rated value is nominal while the measured value is the actual, real-time value.

Experiment 2

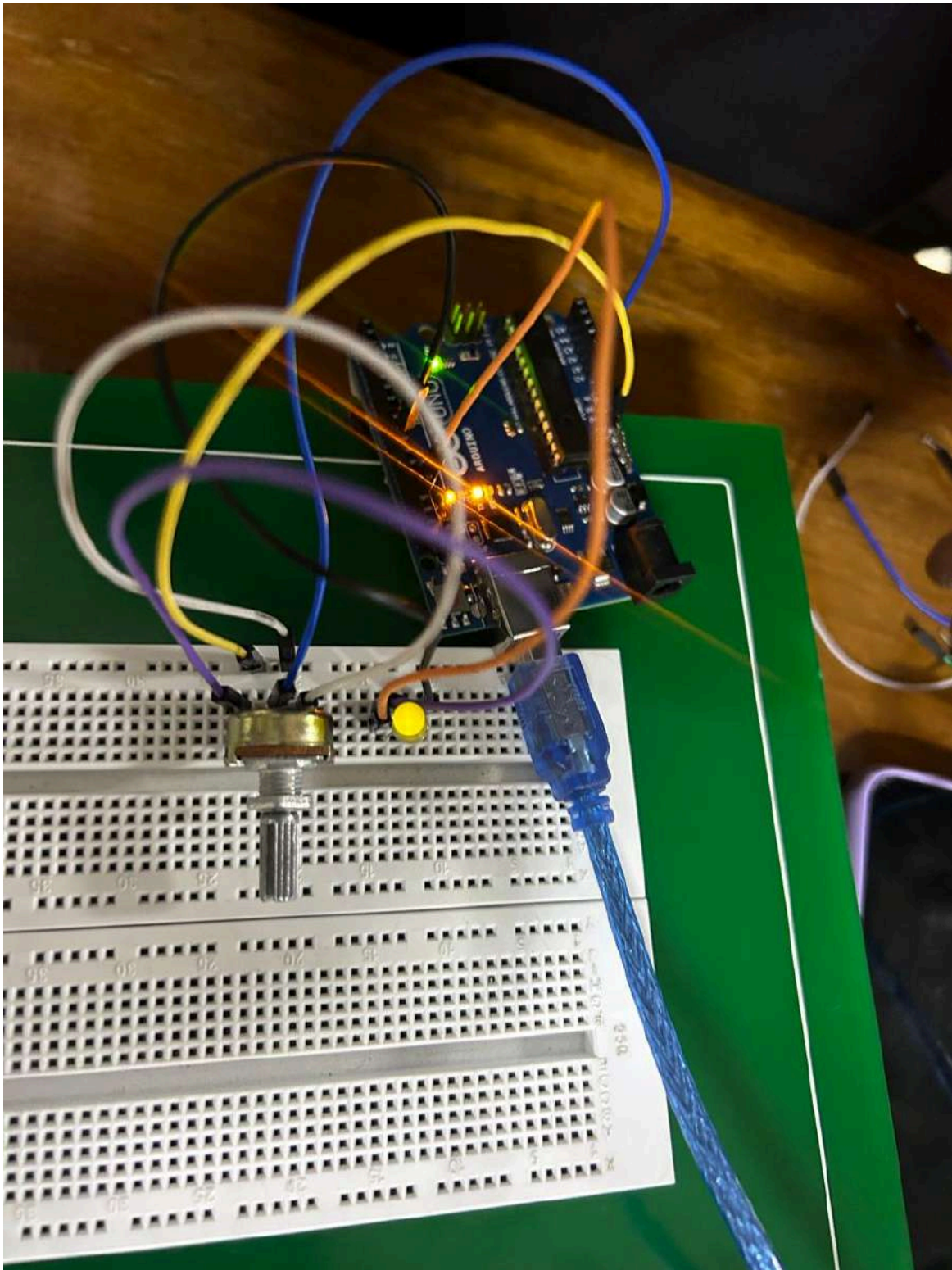
1. Physical Implementation



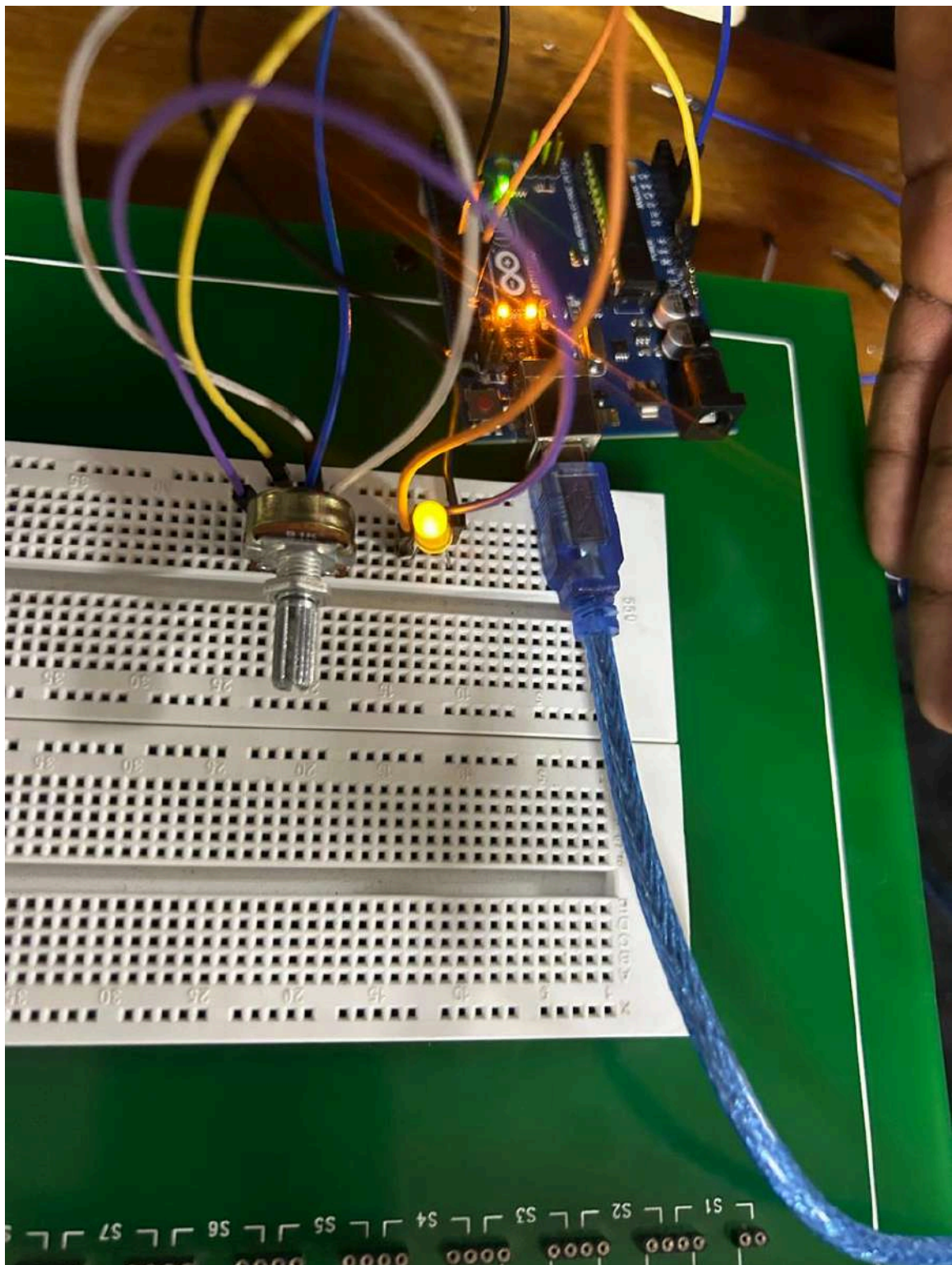
3. Voltage passing through = 4.778 V



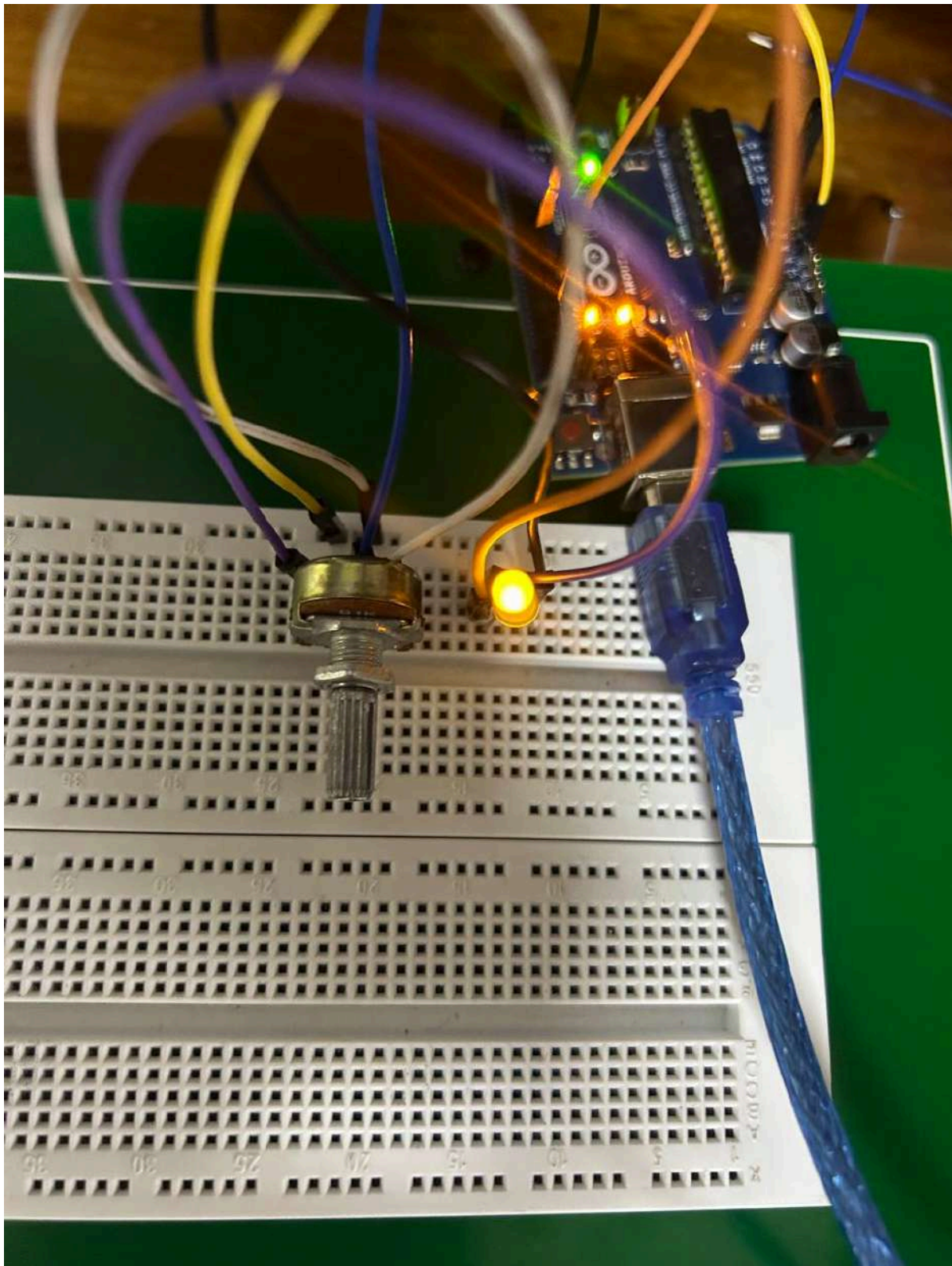
4. Pulse Width Modulation 25%



50%

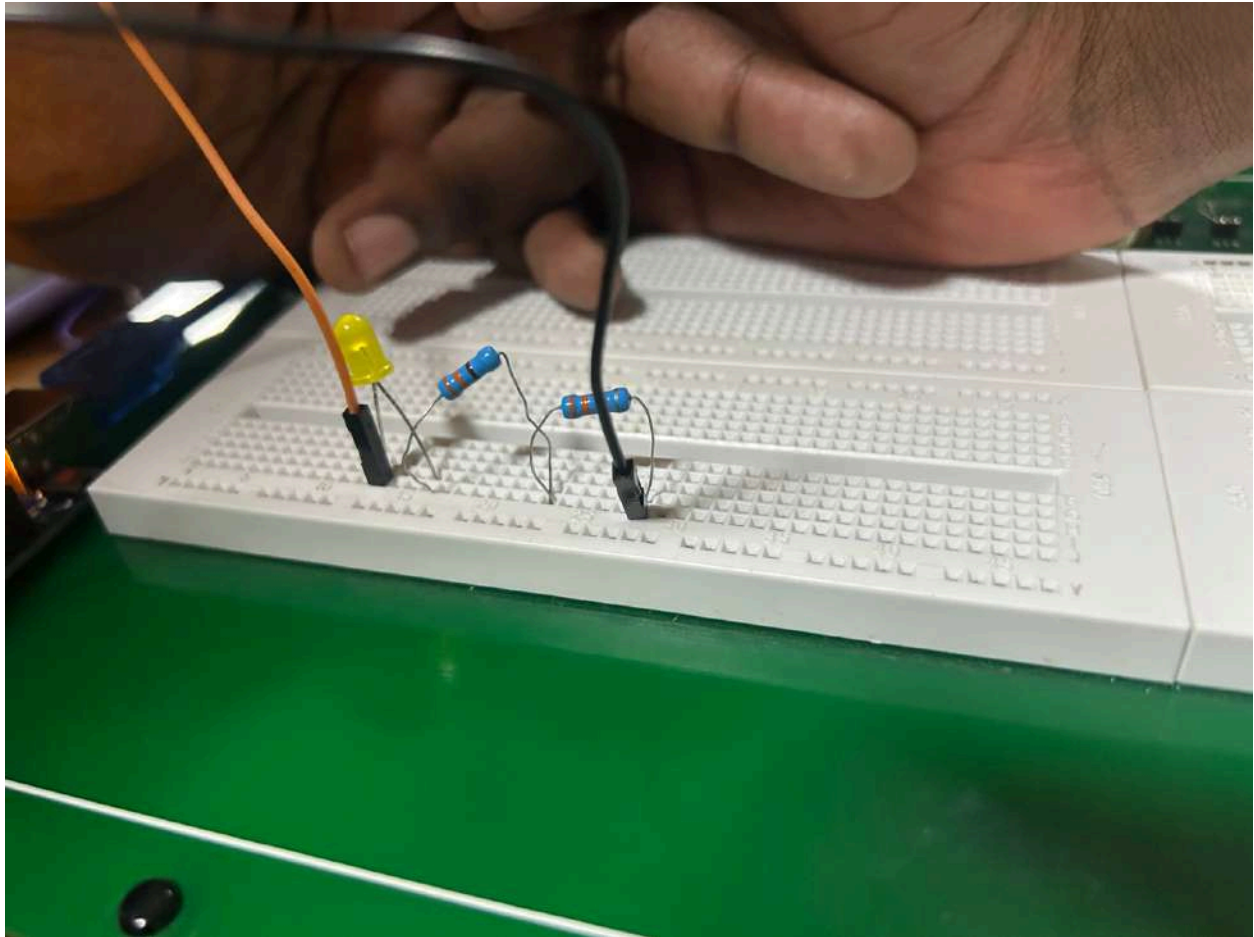


100%

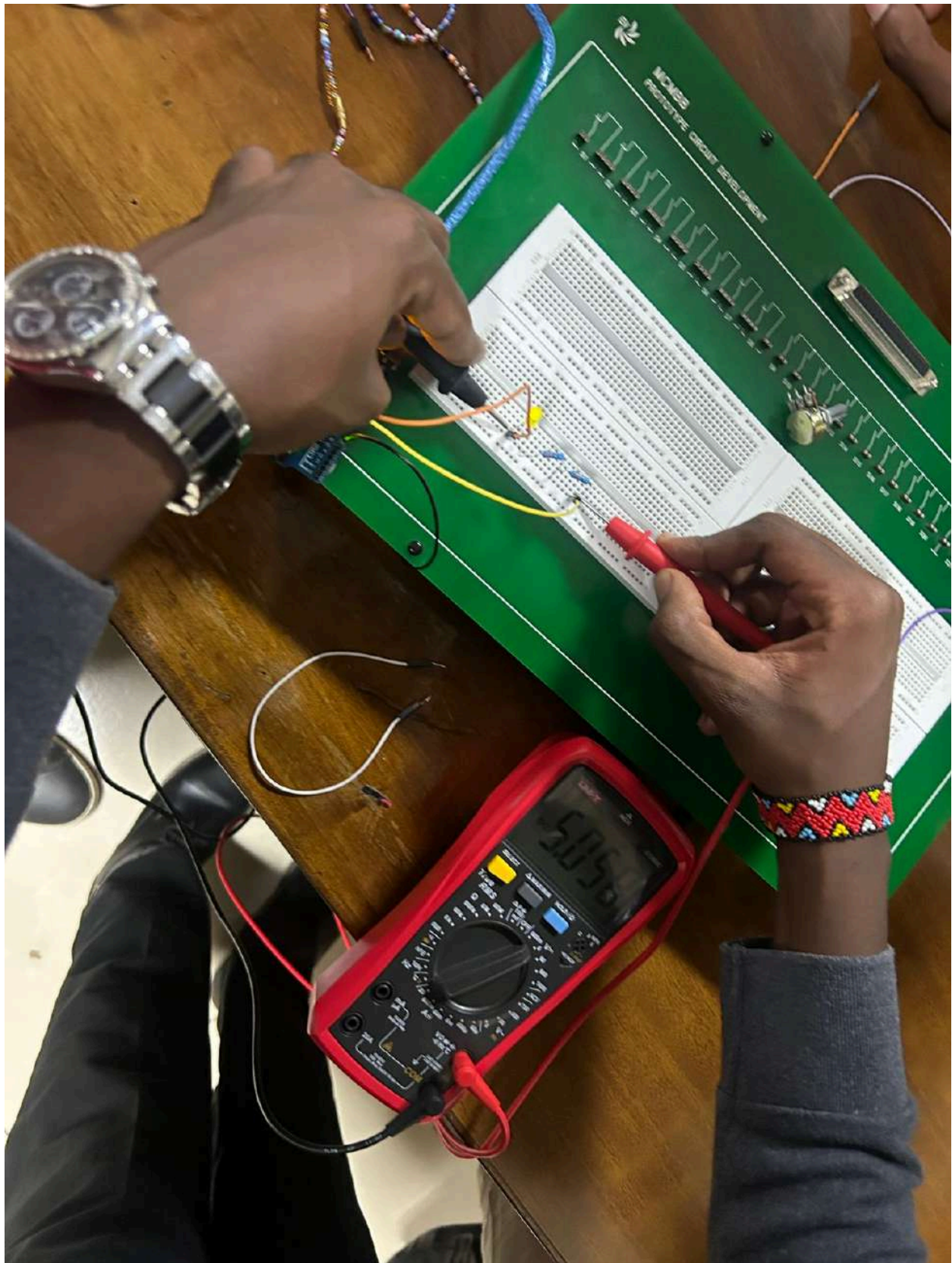


Experiment 3

1. Physical Implementation

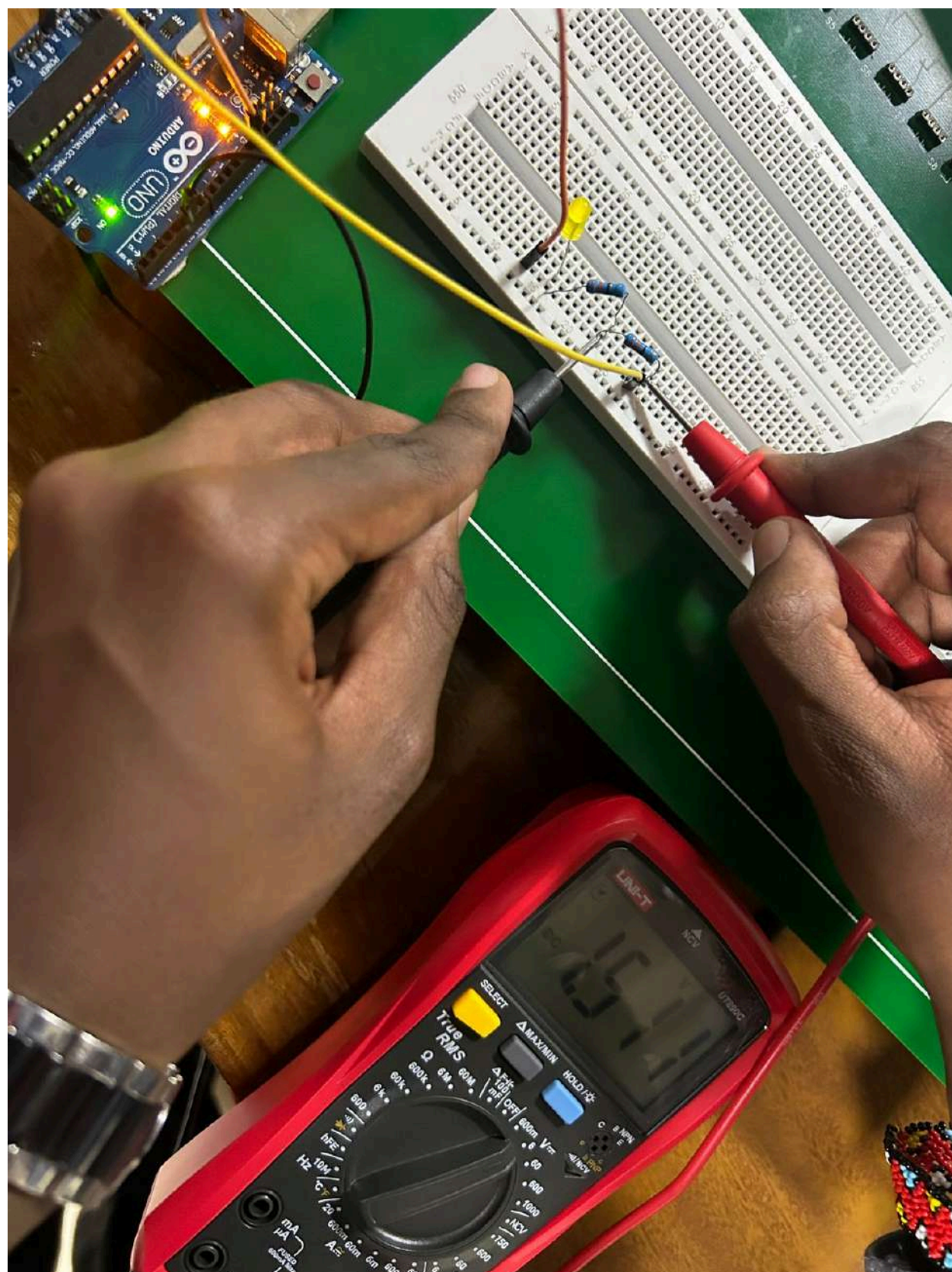


3. Voltage across Circuit = 5.056 V

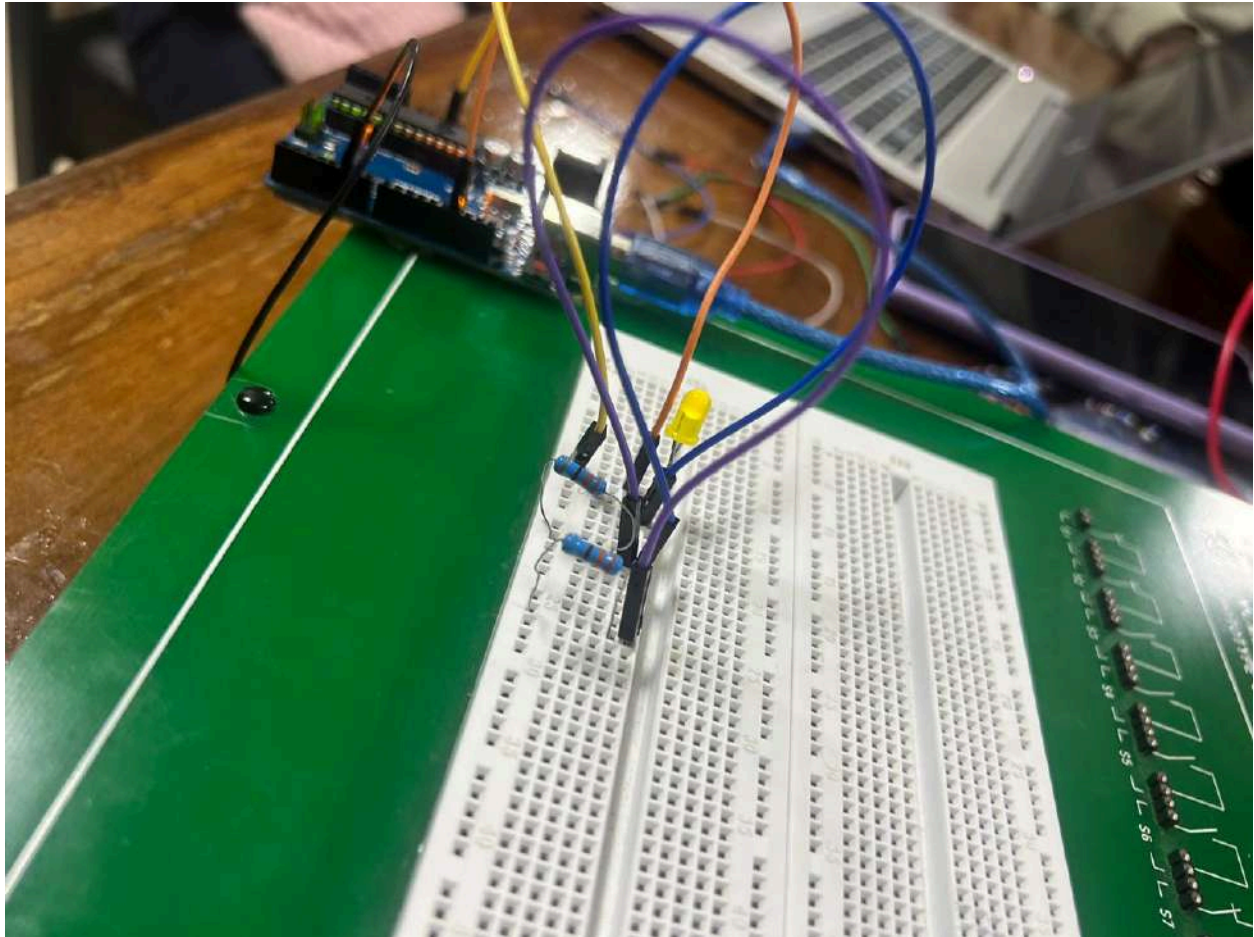


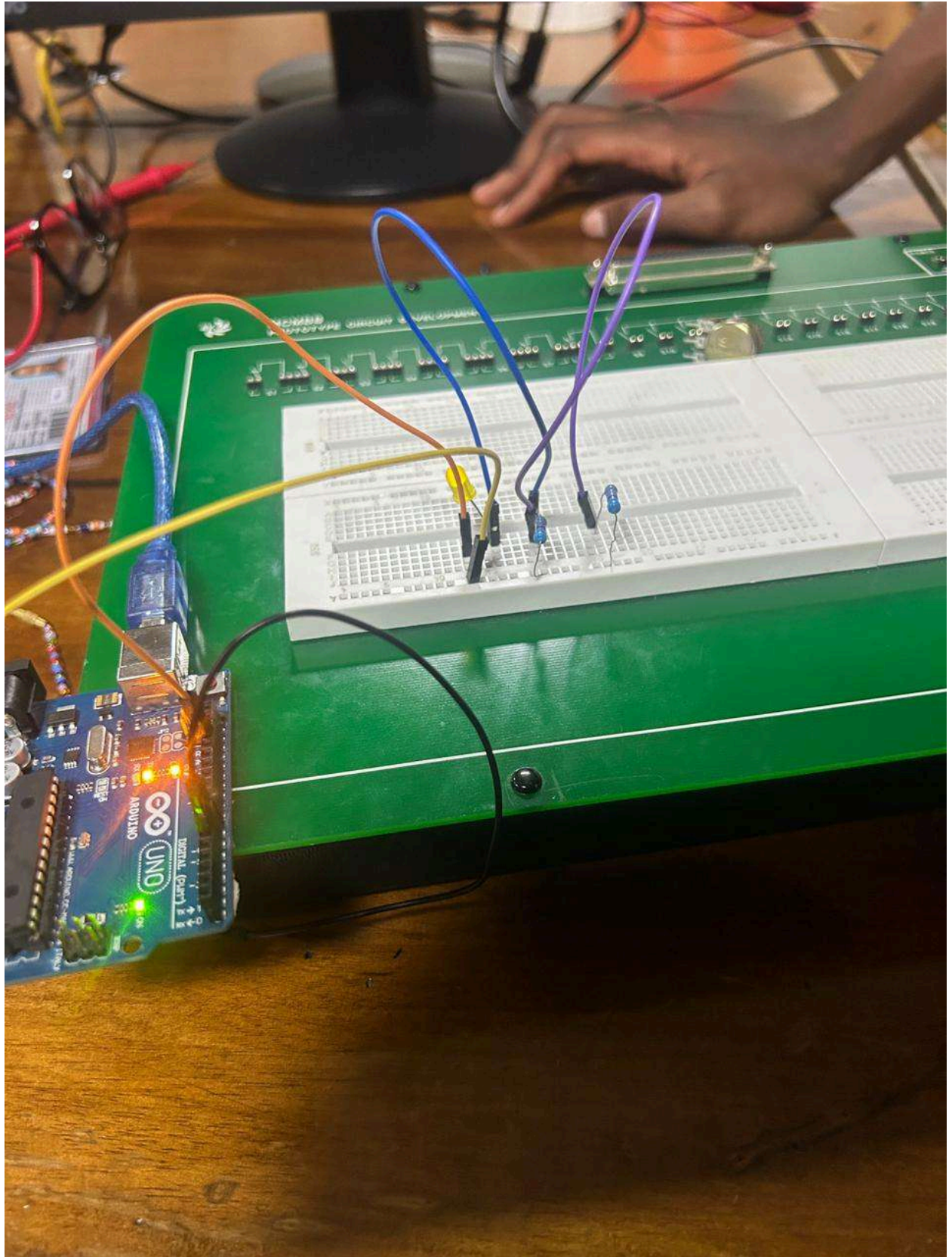
Voltage across each resistor = 1.508 V & 1.511 V



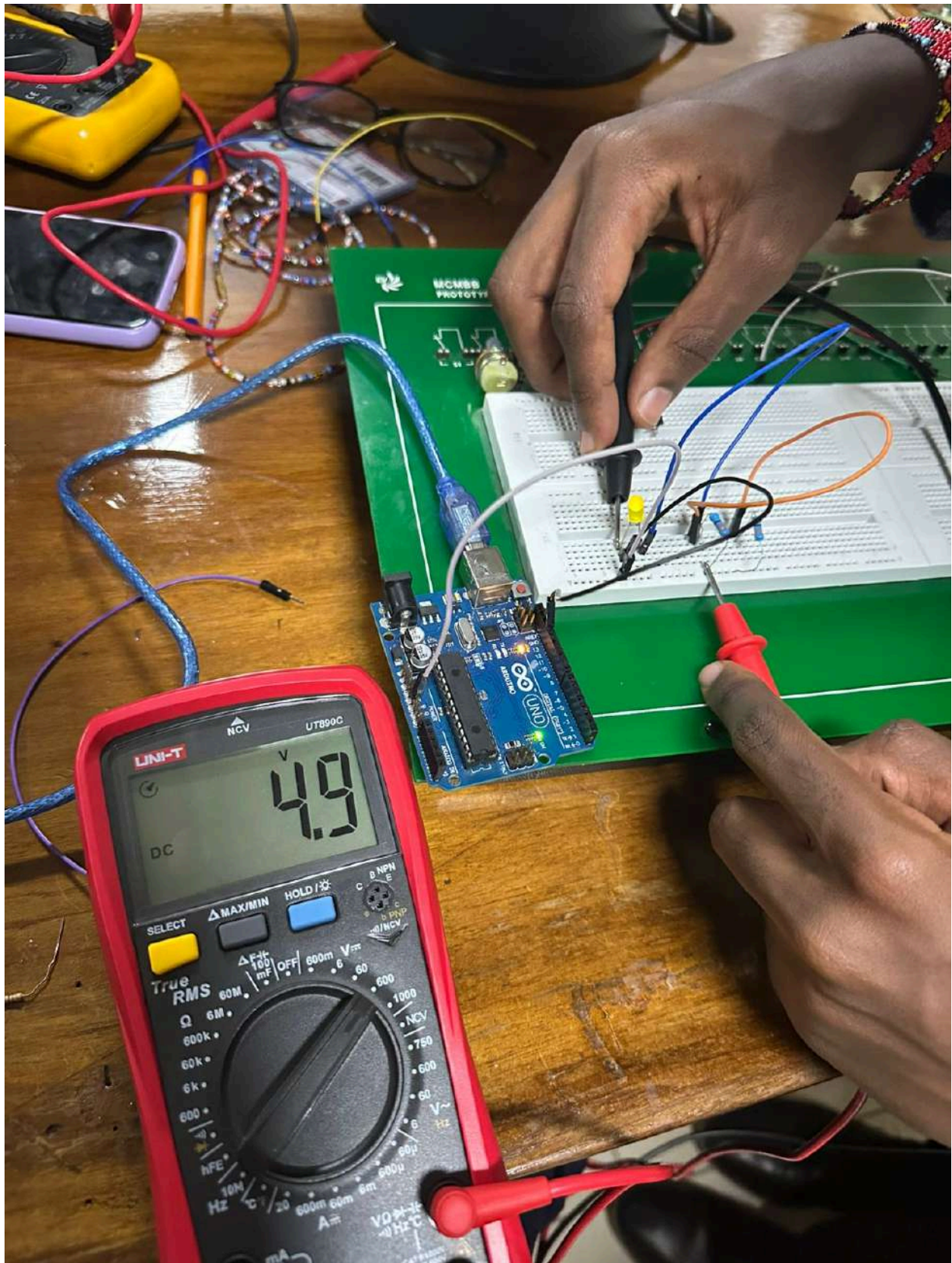


Experiment 4
1. Physical Implementation

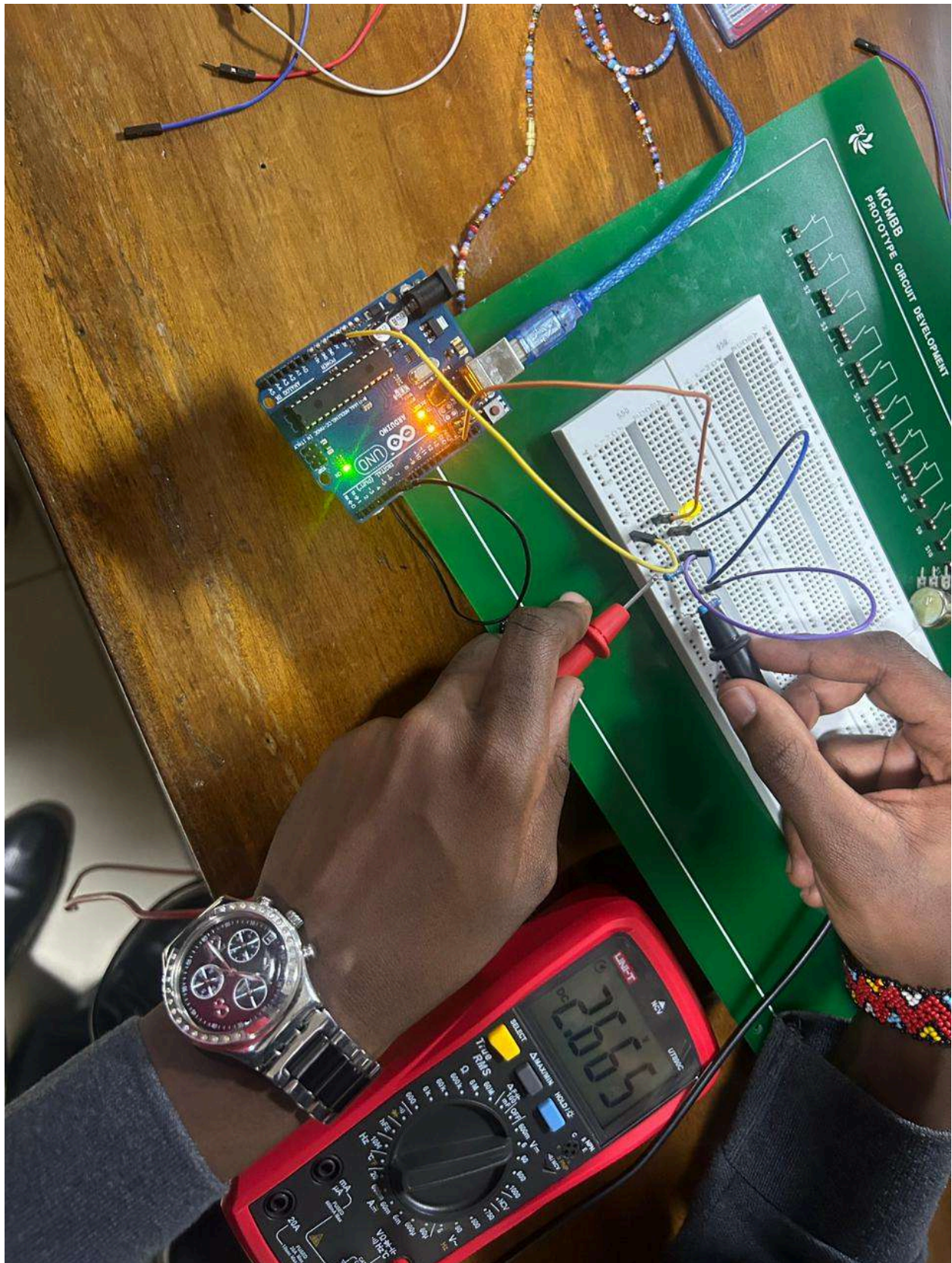


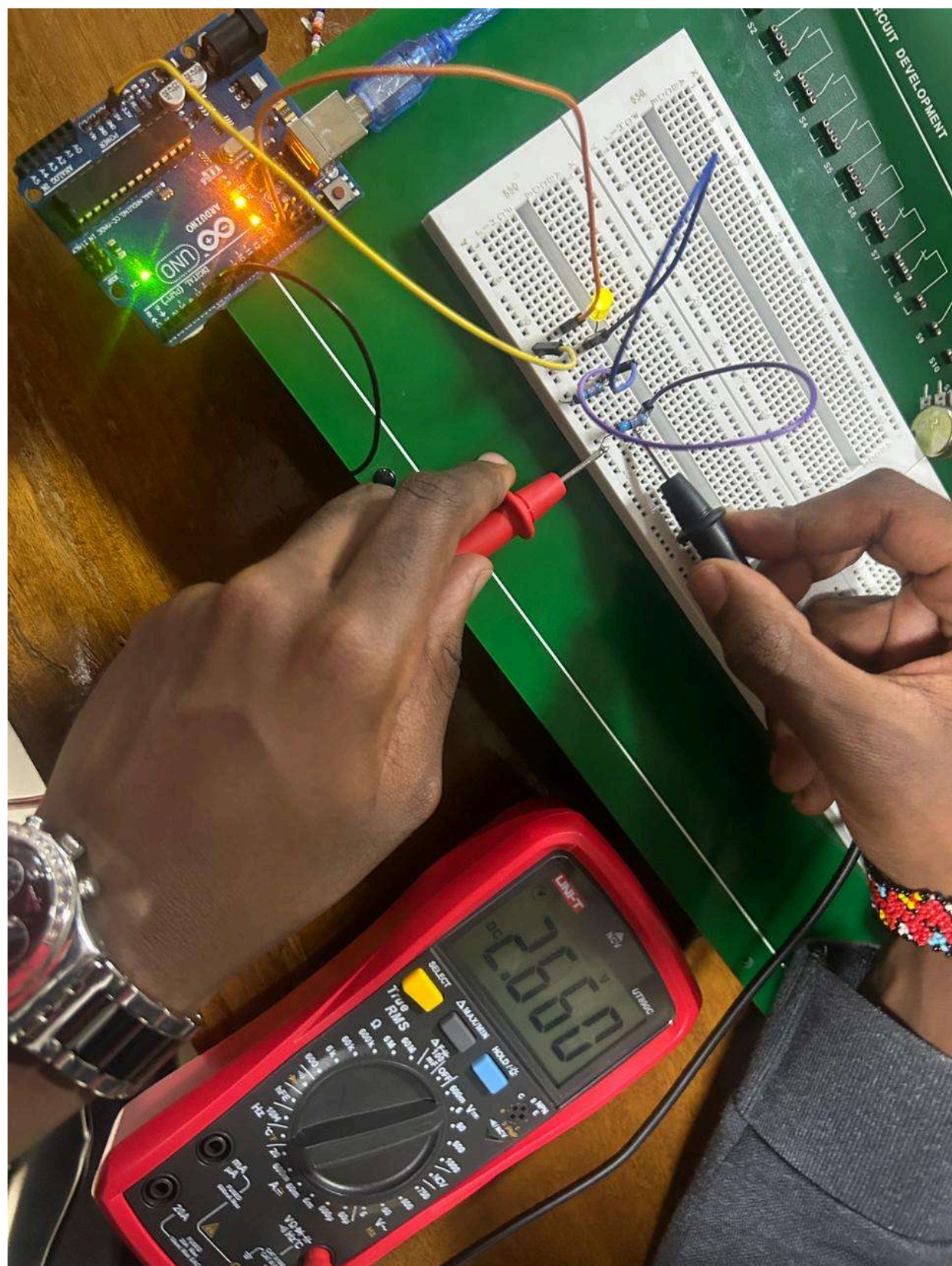


3. Voltage across the circuit = 4.9 V



Voltage across each resistor = 2.665 V & 2.660 V





Use Ohm's Law to estimate the current passed at each resistor.

Resistor 1

$$V = 2.665 \text{ V}$$

$$R = 326.6 \, \Omega$$

$$I = V/R = 2.665/326.6$$

$$\text{Current} = 0.008159828536436 \text{ A}$$

Resistor 2

$$V = 2.660 \text{ V}$$

$$R = 326.6 \, \Omega$$

$$I = V/R = 2.660/326.6$$

$$\text{Current} = 0.0081445192896509 \text{ A}$$